

Calendar Anomalies in the Colombo Stock Exchange: Evidence from Market-Wide and Sector-Level Analysis

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Abstract. Calendar anomalies remain a persistent empirical challenge to the efficient market hypothesis, particularly in frontier markets whose institutional structure and investor composition differ materially from developed economies. This study conducts a systematic investigation of calendar anomalies in the Colombo Stock Exchange (CSE), combining a 26-year market-wide analysis of the All Share Price Index (5,437 daily returns, January 2000–April 2026) with a decade-long sector-level analysis spanning all 20 CSE sectoral sub-indices (40,328 sector-day observations, August 2016–April 2026). Using non-parametric Kruskal–Wallis and Mann–Whitney U tests suited to the fat-tailed, non-normal return distributions observed (kurtosis = 32.098), with primary conclusions additionally confirmed by heteroskedasticity- and autocorrelation-consistent (HAC) regression, the analysis examines month-of-year, turn-of-month, day-of-week, and pre-holiday/Poya-day effects, and evaluates regime-conditionality across pre-crisis, crisis (March 2022–November 2023), and post-crisis phases. The day-of-week effect is highly significant market-wide ($H = 90.57$, $p < 0.0001$) and, after Benjamini–Hochberg correction for the 60 sector-level tests conducted, remains significant in 17 of 20 sectors; the month-of-year effect is significant ($H = 55.99$, $p < 0.0001$), with the pattern largely driven by lower March returns; however, the April-versus-other-month comparison is not statistically significant ($t = 0.467$, $p = 0.6405$); and the pre-holiday effect is the most robust anomaly identified ($t = 2.785$, $p = 0.006$). Hierarchical clustering on a 17-dimensional calendar signature matrix identifies four sector clusters with distinct calendar sensitivity. Monday underperformance was more pronounced during the 2022 crisis at the level of raw means, though a formal HAC-robust weekday-by-regime interaction test did not find this intensification to be statistically significant ($p = 0.310$), indicating the effect should be interpreted with caution. The findings carry direct implications for market surveillance, Poya-closure scheduling, and calendar-aware forecasting in frontier markets.

Keywords: Calendar Anomalies · Hierarchical Clustering · Non-Parametric Testing · Colombo Stock Exchange · Data-Driven Decision Support · Frontier Markets.

1 Introduction

1.1 Background and Motivation

Financial markets are generally assumed to incorporate available information efficiently, yet empirical research has documented recurring return patterns associated with particular days, months, trading cycles, and market closures. These calendar anomalies challenge strict forms of market efficiency and have been linked to behavioural, institutional, and market-microstructure factors. Evidence from emerging and frontier markets suggests that such effects may be especially relevant where liquidity constraints, investor composition, and institutional conditions differ from those of mature markets.

Sri Lanka provides a distinctive setting for examining these effects. The Colombo Stock Exchange (CSE) operates within a frontier-market environment characterised by macroeconomic volatility and a distinctive institutional calendar, including mandatory closures on Buddhist full-moon (Poya) days and an April fiscal-year transition. Although previous studies provide evidence of calendar effects in Sri Lanka, systematic evidence covering sectoral heterogeneity, regime dependence, and cross-sector patterns remains limited [7,12].

1.2 Research Problem and Objectives

Where calendar anomalies remain undocumented, capital is potentially misallocated and surveillance frameworks lack an empirical baseline against which abnormal trading activity can be judged. This study addresses that gap by analysing calendar effects in the CSE at the market and sector level, guided by three questions: do persistent anomalies exist and do sectors cluster into groups sharing similar calendar signatures? Are these effects stable across macroeconomic regimes, including the 2022 sovereign debt crisis? What do the observed patterns reveal about market efficiency and capital market regulation in a frontier economy?

1.3 Contribution and Structure of the Paper

The paper makes four contributions: it provides a comprehensive market- and sector-level analysis of CSE calendar anomalies; identifies heterogeneous sectoral sensitivity through hierarchical clustering; examines regime dependence using a formally tested weekday-by-regime interaction; and translates the findings into implications for market surveillance, Poya-closure scheduling, and calendar-aware forecasting. The remainder of the paper presents the literature, methodology, empirical findings, discussion, and conclusions.

2 Literature Review and Theoretical Background

2.1 Market Efficiency, Behavioural Finance, and Calendar Effects

Calendar effects lie at the intersection of market efficiency and behavioural finance. Under the efficient market hypothesis, available information should be

reflected in prices and systematic abnormal returns should not persist [5]. Nevertheless, day-of-week, month-of-year, turn-of-month, and holiday effects have been documented across financial markets [2–4,6,10]. Behavioural biases, portfolio rebalancing, market microstructure, liquidity conditions, and institutional factors have been proposed as explanations [14–16]. These mechanisms may be particularly relevant in emerging and frontier markets, where institutional and trading conditions differ from those of mature markets [1,8,13].

2.2 The Sri Lankan Context and Methodological Considerations

Sri Lanka provides a distinctive institutional setting because of its April fiscal-year transition, mandatory Buddhist full-moon (Poya) closures, and changing market conditions. Existing evidence indicates calendar effects in the CSE, but sector-level and regime-sensitive evidence remains limited [7,12]. Because aggregate results may conceal industry heterogeneity, this study examines all 20 sectoral indices. Kruskal–Wallis and Mann–Whitney tests provide the primary non-parametric analysis, Ward’s method is used for hierarchical clustering [9,11,17], and HAC (Newey–West) regression provides a robustness check for serial dependence and heteroskedasticity.

3 Data and Methodology

3.1 Data Sources and Sample Period

The market-wide analysis uses 5,437 daily All Share Price Index (ASPI) returns spanning January 2000 to April 2026, a 26-year window encompassing five major macroeconomic shock episodes: the 2001 recession, the 2008 global financial crisis, the 2019 Easter attack, the 2020 COVID-19 pandemic, and the 2022 LKR sovereign debt crisis. ASPI daily returns range from -12.97% to $+20.07\%$, with a kurtosis of 32.098, confirming fat-tailed, non-normal return distributions that motivate the use of non-parametric methods throughout. The sector-level analysis extends this to all 20 CSE sectoral sub-indices from August 2016 to April 2026, yielding 40,328 sector-day observations. The full sample is additionally stratified into pre-crisis (before March 2022), crisis (March 2022–November 2023), and post-crisis (December 2023 onwards) regimes.

Data quality is a material consideration for empirical work of this kind, particularly in a frontier market where reporting standards and market microstructure differ from those of developed exchanges. The completeness checks and preprocessing steps applied before the datasets described above were used in the analysis are outlined below.

Missing observations were classified as scheduled non-trading days, such as weekends and Poya closures, or other gaps requiring review. Returns were calculated from consecutive trading days, while records with missing prices, volume, or turnover were excluded rather than imputed. Extreme observations associated with genuine market events were retained, whereas implausible data-entry values were removed.

Data Completeness. After these exclusions, the market-wide series retained 5,437 daily observations and the sector-level panel retained 40,328 sector-day observations. Completeness was assessed as the ratio of actual to theoretically expected sector-days, where the theoretical maximum accounts for each sector’s own observed launch date rather than assuming all 20 sectors were active throughout the full study window — most sectors began on 2 August 2016, Real Estate Management & Development on 18 September 2016, and Software & Services, the most recently introduced CSE sub-index, only from 24 December 2021. On this basis, the theoretical maximum is 49,238 sector-days, against which the 40,328 retained observations represent 81.9% completeness. The residual gap is consistent with scheduled CSE closures (Poya days and public holidays, which are not captured by a simple business-day calendar) together with a small number of raw records removed during deduplication of sector-date collisions in the source export; this figure is therefore reported as a conservative completeness estimate rather than as a qualitative claim.

3.2 Calendar Variables and Empirical Strategy

The paper examines five calendar dimensions: day-of-week, month-of-year (particularly the March discount ahead of Sri Lanka’s April fiscal-year start), turn-of-month, pre-holiday/Poya-day, and regime-sensitive effects across market-stress and recovery periods. The empirical analysis proceeds from the broad market index to individual sectors and days, using Kruskal–Wallis H tests [9] as the primary framework, supplemented by Mann–Whitney U pairwise comparisons [11] and t-tests where applicable, with all primary conclusions re-estimated using HAC-robust (Newey–West, 5-lag) regression as a check. Because Table 1 reports 60 sector-level tests (20 sectors $\times$ 3 dimensions), Benjamini–Hochberg false discovery rate correction was applied, with corrected counts reported alongside nominal results (Sect. 4). Regime-conditionality was tested via a formal weekday $\times$ regime interaction model (HAC-robust OLS), rather than relying on a comparison of raw regime means alone. Cross-sector pattern recognition employs hierarchical clustering with Ward linkage [17] on a 17-dimensional calendar signature matrix (five day-of-week and twelve monthly demeaned effects per sector) — features also relevant to return forecasting, consistent with the broader literature on hybrid, feature-enriched frameworks [18], a direction pursued in companion work using an LSTM architecture enriched with CSE-specific calendar features.

4 Empirical Findings

4.1 Month-of-Year and Turn-of-Month Effects

The month-of-year effect was significant at the market level (Kruskal–Wallis H = 55.99, $p < 0.0001$), with March emerging as the weakest universal month (-0.366% mean deviation across all 20 sectors), visible in 14 of 20 sectors. Figure 1 shows March returns are markedly negative and followed by a recovery from

April, consistent with end-of-fiscal-year portfolio liquidation. A direct comparison of April against all other months was not statistically significant in isolation ($t = 0.467$, $p = 0.6405$), indicating that the April effect is better understood as a recovery from the March trough rather than an anomaly confined to April itself. The turn-of-month effect was not significant at the market-wide level ($t = -0.759$, $p = 0.448$) and, after Benjamini–Hochberg correction, was not significant in any sector. HAC-regression estimates were consistent with these conclusions across all four effects (Sect. 3.2).

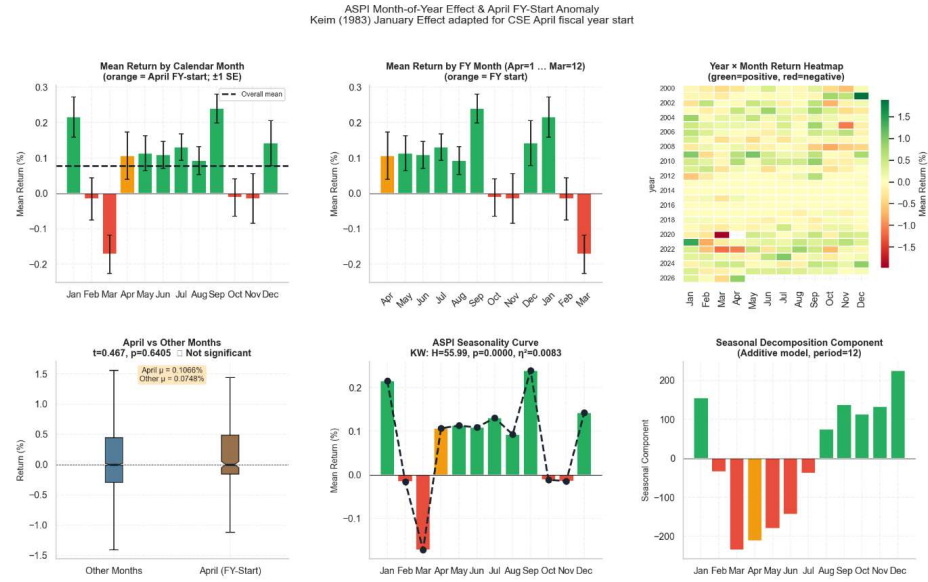


Fig. 1. ASPI month-of-year effect and April fiscal-year-start anomaly. Mean returns by calendar month and fiscal-year month show March as the weakest month and a recovery from April.

4.2 Day-of-Week Effects

The day-of-week effect was highly significant at the market-wide level (Kruskal–Wallis $H = 90.57$, $p < 0.0001$), with Monday returns systematically below the market mean (-0.069%), confirmed by Mann–Whitney U ($p < 0.0001$) and by HAC-robust regression, while Thursday and Friday returns were the strongest days ($+0.226\%$ and $+0.203\%$ respectively). Figure 2 illustrates the pattern together with the year-by-year Monday effect, which was negative in 21 of the last 27 years. This modified Monday effect appears amplified by low intraday liquidity and weekend news being absorbed in a thin-trading environment. At the sector level, the effect was significant in 19 of 20 sectors under nominal testing

($p < 0.05$); after Benjamini–Hochberg correction for the 60 sector-level tests reported in Table 1, the effect remains significant in 17 of 20 sectors, with Monday negative and Friday positive in all 20 sectors (Fig. 6, Table 1).

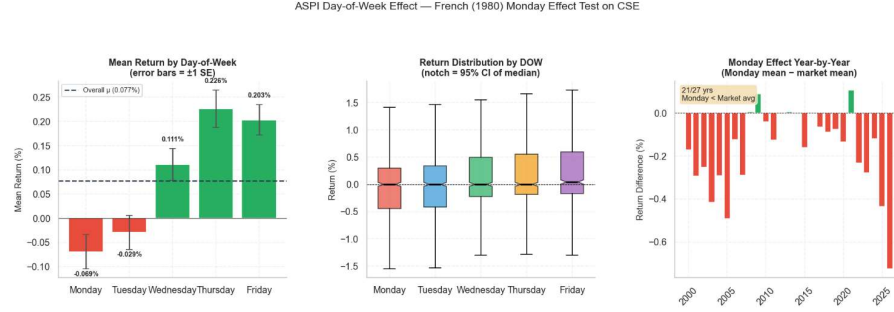


Fig. 2. ASPI day-of-week effect. Mean returns by trading day show significant Monday underperformance and Wednesday–Friday outperformance, confirmed by Kruskal–Wallis ($H = 90.57$, $p < 0.0001$).

4.3 Pre-Holiday and Poya-Day Effects

The pre-holiday effect was the most robust finding of the study ($t = 2.785$, $p = 0.006$), with pre-holiday mean returns of $+0.321\%$ compared to $+0.069\%$ on ordinary trading days. Figure 3 shows that returns rise systematically as a Poya day approaches, consistent with short-covering pressure accumulating before each full-moon closure. Ranking all calendar anomaly groups by mean return places the pre-holiday effect above every other calendar dimension examined, ahead of Thursday ($+0.226\%$), Friday ($+0.203\%$), and the April fiscal-year effect ($+0.107\%$), confirming it as the single most economically meaningful calendar anomaly identified in the CSE.

4.4 Cross-Sector Structure and Clustering

Hierarchical clustering on sector calendar signatures (17-dimensional day-of-week and monthly vectors, Ward linkage) identified four distinct groups (Fig. 4). Software & Services, Health Care, Telecommunication Services, and Insurance formed the most calendar-sensitive cluster, while a larger cluster of cyclically linked sectors — including Banks, Diversified Financials, Capital Goods, and Materials — shared a more muted seasonal signature. Insurance was the sole sector significant across all three calendar dimensions examined under nominal testing (day-of-week, monthly, and turn-of-month; Table 1). The cross-sector monthly heatmap (Fig. 5) and day-of-week heatmap (Fig. 6) show that the March discount and the Monday/Friday asymmetry are broad-based across the

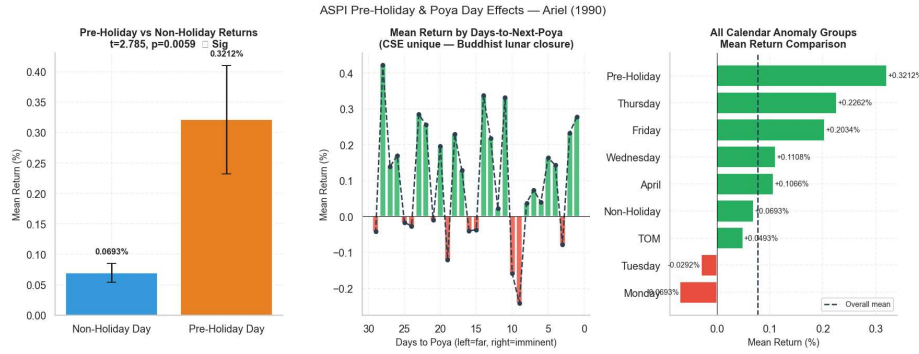


Fig. 3. ASPI pre-holiday and Poya-day effects. Pre-holiday returns (+0.321%) significantly exceed non-holiday returns (+0.069%, $t = 2.785$, $p = 0.006$), with returns rising as Poya day approaches — the most robust calendar anomaly identified.

exchange rather than confined to a handful of sectors; after Benjamini–Hochberg correction, 17 of 20 sectors remain significant on DOW, 12 of 20 on Monthly, and 0 of 20 on TOM (Table 1).

4.5 Regime Dependence

Regime analysis examined whether calendar anomalies are static features of the market. At the level of raw means, Monday returns worsened from -0.054% in the pre-crisis period to -0.247% during the 2022–2023 crisis, while cross-sector volatility more than doubled over the same window. However, a formal weekday $\times$ regime interaction model (HAC-robust OLS, 5-lag Newey–West standard errors) found this apparent intensification was not statistically significant (coefficient = -0.207 , 95% CI $[-0.606, 0.192]$, $p = 0.310$), likely reflecting the limited number of crisis-period Mondays ($n \approx 80$). The descriptive pattern is therefore consistent with calendar-linked behavioural and liquidity frictions intensifying during periods of macroeconomic uncertainty, but this intensification should be treated as suggestive rather than statistically confirmed, and it motivates formally testing (rather than only describing) regime-conditionality in future calendar-effect research.

5 Discussion

5.1 Market Efficiency and Institutional Context

The persistence of day-of-week effects across 17 of 20 sectors after multiple-testing correction, together with the robust pre-holiday effect, indicates that calendar-linked return patterns remain present in the CSE. These findings are consistent with behavioural explanations of asset pricing [14,15]. The Poya closure calendar, April fiscal-year transition, and local investor composition further

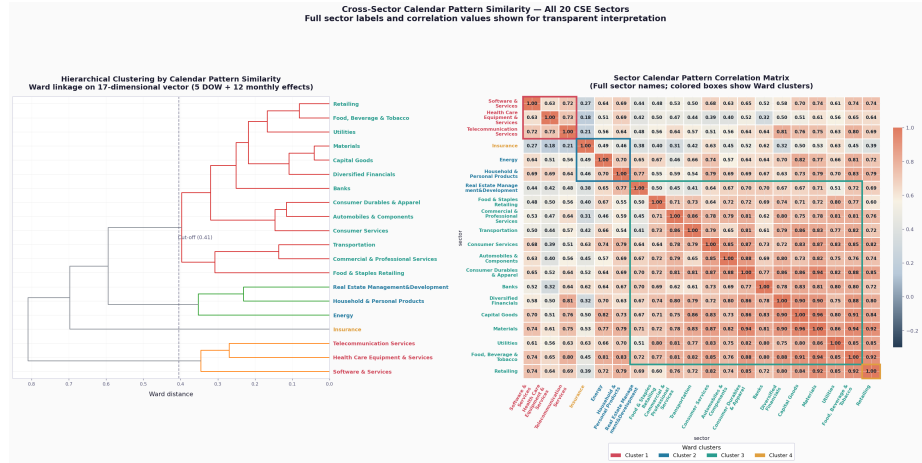


Fig. 4. Hierarchical clustering (Ward linkage) of all 20 CSE sectors by calendar pattern similarity, identifying four distinct clusters of sectors sharing similar day-of-week and month-of-year calendar signatures.

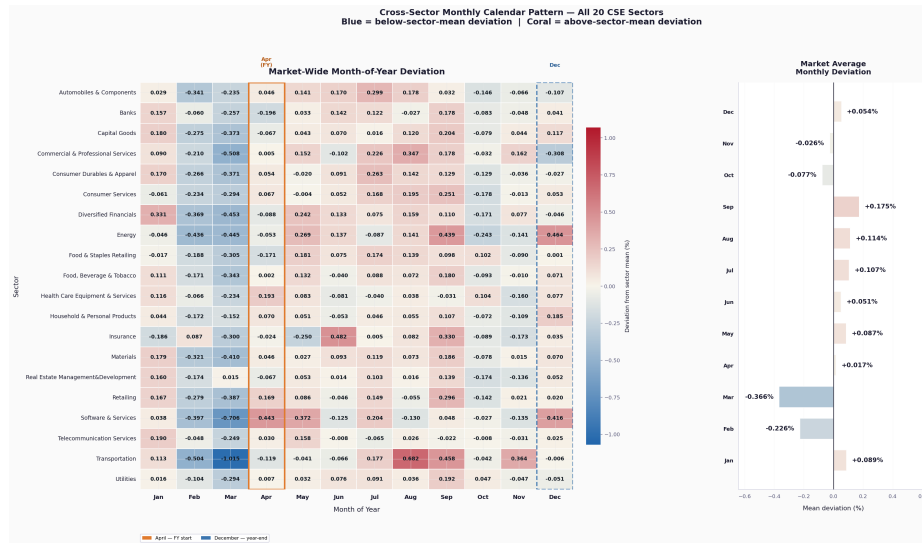


Fig. 5. Cross-sector monthly calendar effect summary across all 20 CSE sectors.

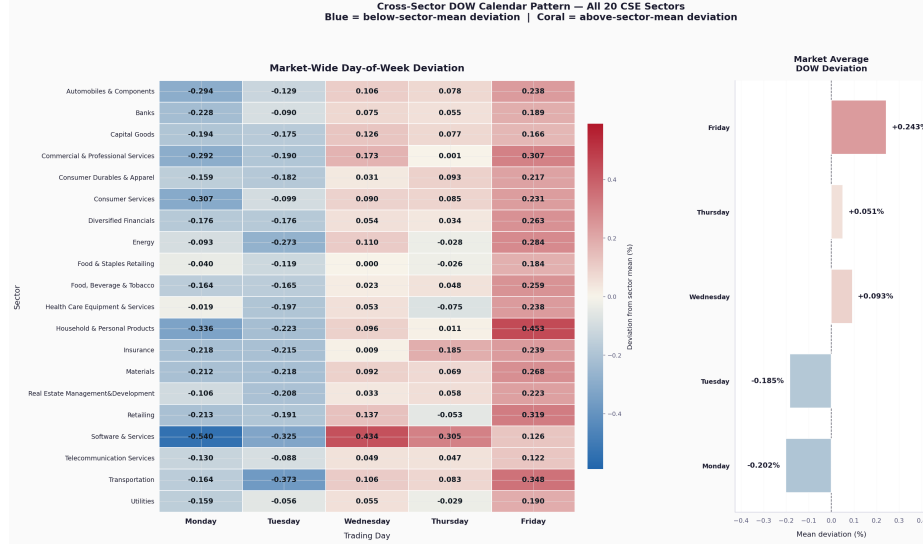


Fig. 6. Cross-sector day-of-week calendar effect summary across all 20 CSE sectors.

Table 1. Cross-sector calendar effect significance: nominal p-values and Benjamini–Hochberg adjusted q-values (bold = significant at $q < 0.05$) by dimension, across the 60 sector-level tests (20 sectors $\times$ 3 dimensions). DOW = day-of-week, TOM = turn-of-month. After correction, 17/20 sectors remain significant on DOW, 12/20 on Monthly, and 0/20 on TOM.

Sector	DOW		Monthly		TOM	
	<i>p</i>	<i>q</i>	<i>p</i>	<i>q</i>	<i>p</i>	<i>q</i>
Insurance	0.0000	0.0000	0.0132	0.0344	0.0341	0.0639
Utilities	0.0274	0.0530	0.0185	0.0409	0.7862	0.8577
Retailing	0.0002	0.0011	0.0187	0.0409	0.1478	0.2274
Capital Goods	0.0000	0.0000	0.0124	0.0338	0.6283	0.7693
Commercial & Prof. Services	0.0218	0.0451	0.0118	0.0337	0.1442	0.2274
Consumer Durables & Apparel	0.0004	0.0017	0.0005	0.0020	0.6128	0.7660
Consumer Services	0.0000	0.0000	0.0085	0.0255	0.3948	0.5150
Diversified Financials	0.0000	0.0000	0.0058	0.0183	0.7601	0.8577
Energy	0.0145	0.0350	0.0146	0.0350	0.2968	0.4141
Transportation	0.0015	0.0053	0.0004	0.0017	0.7815	0.8577
Food, Beverage & Tobacco	0.0000	0.0000	0.0009	0.0034	0.3096	0.4222
Materials	0.0000	0.0000	0.0004	0.0017	0.3665	0.4887
Telecommunication Services	0.0191	0.0409	0.1886	0.2760	0.6431	0.7717
Software & Services	0.0001	0.0006	0.0556	0.1011	0.9727	0.9727
Automobiles & Components	0.0028	0.0093	0.1460	0.2274	0.0652	0.1151
Real Estate Mgmt. & Dev.	0.0000	0.0000	0.1525	0.2288	0.6894	0.8054
Household & Personal Products	0.0000	0.0000	0.8109	0.8688	0.8625	0.8771
Banks	0.0000	0.0000	0.1974	0.2820	0.8411	0.8771
Health Care Equip. & Services	0.0259	0.0518	0.6980	0.8054	0.6033	0.7660
Food & Staples Retailing	0.0837	0.1435	0.1080	0.1800	0.8534	0.8771

distinguish the CSE from developed markets and suggest that institutional context is important when interpreting calendar anomalies.

5.2 Sectoral, Policy, and Forecasting Implications

The four sector clusters provide a basis for calendar-aware portfolio and forecasting strategies. Market surveillance could incorporate calendar-adjusted benchmarks, particularly for Monday and pre-holiday activity, while the CSE and relevant regulators could consider calendar effects when evaluating unusual trading patterns. The non-significant crisis interaction suggests that regime-based monitoring should remain precautionary. The clustering results also support using shared calendar features in sector-level forecasting rather than developing completely independent specifications for each sector [18].

6 Conclusion

6.1 Summary of Contributions

This study provides market- and sector-level evidence of calendar anomalies in the CSE using 26 years of market-wide data and approximately a decade of sector observations across all 20 industries. Day-of-week and pre-holiday effects are particularly persistent, while turn-of-month effects are largely absent. After Benjamini–Hochberg correction, day-of-week effects remain significant in 17 of 20 sectors, and hierarchical clustering identifies four groups with distinct calendar signatures. Although Monday underperformance was descriptively stronger during the 2022 crisis, the HAC-robust weekday-by-regime interaction was not statistically significant. The findings highlight the importance of institutional and sectoral context in frontier-market calendar effects and support applications in market surveillance and calendar-aware forecasting.

6.2 Limitations and Future Research

The sector classification follows the CSE’s standard groupings and therefore may not capture firm-level heterogeneity. Future work could extend the analysis to individual firms and use formal regime-switching models to identify changing calendar effects. The results could also be incorporated into multivariate forecasting models and compared with other frontier markets in South Asia to assess their generalisability.

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